

SUKKUR IBA UNIVERSITY

Department of Computer Science

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Real-Time Player Confidence & Mood Analysis System

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Problem Statement

In competitive gaming and sports, a player's emotional state plays a crucial role in their performance. Currently, there is no automated, real-time system that can read a player's facial expressions and assess their confidence level or emotional mood during gameplay. Coaches, analysts, and team managers rely on subjective observation, which is inconsistent and often delayed. This project aims to address that gap by building an AI-powered system that automatically detects and classifies player mood in real-time using computer vision.

Proposed Solution

We propose to develop a Real-Time Facial Expression and Mood Detection System that uses a webcam as input to continuously capture a player's face, processes it through a trained Convolutional Neural Network (CNN), and outputs the detected emotional state. The system will follow a simple pipeline:

- **Input:** Live video feed captured through a webcam
- **Face Detection:** Identify and extract the face region from each frame
- **Preprocessing:** Resize, normalize, and convert frames to grayscale
- **Model Inference:** Pass the face through the trained CNN model
- **Output:** Display the detected mood/emotion label in real-time on screen

The final application will be deployed as a Streamlit web app, allowing live demonstration and testing through a simple and user-friendly graphical interface.

Dataset

Name: FER-2013 (Facial Expression Recognition 2013)

Source: Kaggle (publicly available)

Size: ~35,000 labeled grayscale facial images (48x48 pixels)

Classes: 7 emotion categories

Emotion Class	Description
Happy	Player is confident and performing well
Sad	Player is disappointed or losing focus
Angry	Player is frustrated with gameplay
Neutral	Player is calm and composed
Surprised	Player reacts to unexpected in-game events

Fear	Player is anxious or under pressure
Disgust	Player is dissatisfied with outcome

AI Model & Techniques

Model Type: Simple Convolutional Neural Network (CNN)

Framework: TensorFlow / Keras (Python)

The CNN architecture will consist of the following layers:

- Convolutional Layers: For automatic feature extraction from facial images
- MaxPooling Layers: For dimensionality reduction and spatial invariance
- Dropout Layers: To prevent overfitting during training
- Fully Connected (Dense) Layers: For final classification into 7 emotion classes
- Softmax Output Layer: To output probability scores for each emotion

This model was chosen for its simplicity, ease of implementation, and suitability for image classification tasks. CNN is also directly related to topics covered in the AI course lectures.

Tools & Technologies

Category	Technology / Library
Language	Python 3.x
Deep Learning	TensorFlow / Keras
Computer Vision	OpenCV (cv2)
Data Handling	NumPy, Pandas
Dataset	FER-2013 via Kaggle
GUI / Deployment	Streamlit
Visualization	Matplotlib, Seaborn

Expected Outcome

Upon completion, the system will provide a fully functional real-time mood detection application with the following features:

- Live webcam feed with real-time face detection and bounding box
- Instant emotion label displayed on screen (e.g., Happy, Angry, Neutral)
- Confidence percentage shown alongside each prediction
- A Streamlit dashboard for interactive testing and demonstration